Haotian Chen (Chahot)

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LinkedIn

Homepage

Google Scholar

Technical Blog (in Chinese)

Research Interests

Questions — My research in quantum computing is centered on two fundamental questions: (1) How can we establish reliable security guarantees for quantum protocols under realistic, finite-resource and computational constraints? (2) How can such guarantees be translated into practical and verifiable cryptographic constructions?

Directions — To address these questions, my work lies at the intersection of: (1) Quantum Cryptography; (2) Near-Term Quantum Computing; (3) Computational Complexity; (4) Quantum Device Certification and Verification.

Experience

Mar. 2026 — Present

Invisible Technologies, Remote and part-time
AI Trainer (Project-Based)
Overview: Contributed to the evaluation and refinement of Korean- and English-language large language model outputs, with a focus on response quality, linguistic accuracy, and instruction adherence.

Oct. 2024 — Apr. 2025

National University of Singapore, Singapore
Research Assistant, Centre for Quantum Technologies
Overview: Analysed quantum authentication and capability certification under adversarial models. Investigated approaches based on unitary inversion and trap-based techniques for distinguishing honest and malicious behaviours.

Dec. 2023 — Jul. 2024

Publishing House of Electronics Industry, Beijing, China
Independent Author
Overview: Authored a technical textbook on quantum security frameworks, covering the transition from classical cryptography to quantum information-theoretic security. Structured key concepts including superdense coding, quantum teleportation, quantum key distribution and post-quantum cryptography.

Sep. 2021 - Oct. 2023

Seoul National University of Science and Technology, Seoul, South Korea
Ph.D. Scholar, Department of Computer Science and Engineering
Overview: Developed a strong foundation in cybersecurity systems and cryptographic frameworks, with early work spanning IoT security, blockchain, and distributed systems. Subsequently redirected research focus toward quantum security, emphasising the integration of quantum computing with cryptographic models and security analysis.

Experience (continued)

Feb. 2022 — Nov. 2023

Seoul National University of Science and Technology, Seoul, South Korea
Assistant Lab Manager, Ubiquitous Computing and Security Lab
Supported research activities through grant proposal preparation and coordination of research projects. Facilitated project execution and collaboration within the lab.

Education

Sep. 2021 – Nov. 2023

Ph.D.(Incomplete), Seoul National University of Science and Technology. Department of Computer Science and Engineering.
Specialisation: Cybersecurity in IoT
Thesis title: *A Smart Contract and Digital Twin-based Heuristic Multi-Cooperation Scheduling Framework for Smart Manufacturing in IIoT Environments.*
GPA:4.27/4.50 (97.4/100)
Programme concluded after a strategic transition of research focus to quantum computing.

Mar. 2017 – Aug. 2021

Bachelor of Science in Computer Science and Engineering, Seoul National University of Science and Technology. Department of Computer Science and Engineering.
Specialisation: Information Security
Thesis title: *Servers—Client Mini-Network Security Prototype Based on VMWare and GNS3.*
GPA:3.64/4.50 (90.2/100)
Rank: Ranked **10th** among Department of Computer Science and Engineering graduates in the Class of 2021.

Teaching & Academic Services

Mar. 2022 - Sep. 2023

Teaching Assistant at Seoul National University of Science and Technology
Assisted in course administration, student coordination, and exam logistics. Responsible for designing exam questions, grading exams and assignments, and maintaining academic records.

Sep. 2019 - Dec. 2020

Physics Mentor at Seoul National University of Science and Technology.
Provided one-on-one tutoring to undergraduate students, assisting them in mastering fundamental physics concepts and problem-solving techniques.

Skills

Languages: Chinese (Native); English (IELTS 7.0); Korean (TOPIK Lv.6).

Coding: Python, Qiskit, Java, C++, LaTeX.

Software and Tools: Docker, Wireshark, Nmap, Jupyter Notebooks, GNS3, VMware, OpenAI API, Scopus, Web of Science, AI-assisted research tools.

Miscellaneous: Strategy-based games, fitness and swimming, technical blogging, intercultural communication

Recognitions

Scholarships

2017 — 2021 **Scholarship for Outstanding Achievement for Foreigners** for each semester of my undergraduate studies.

2021 — 2023 **Scholarship for Outstanding Achievement for Foreigners** for each semester of my graduate studies.

Best Paper Award(Multiple)

2022
- The 2022 World Congress on Information Technology Applications and Services Jeju, Korea
- The International Conference on Big data, IoT, and Cloud Computing, Jeju, Korea
- The International Conference on Future Information Technology, applications and services, Seoul, Korea
- The International Conference on Computer Science and its Applications, Vientiane, Laos

Certification and Professional Service

2023
- **Alibaba Cloud Developer Community-Blogging Expert**. Alibaba Changfengzhe Program.
- **CSDN-Blogging Expert**. Subscriptions more than 20,000 and total number of visits is about 600,000.
- **Examiner**. Working for the security section of the Korean Information Processing Technology Professional Certification Examination.

Research Publications

Journal Articles

1. **H. Chen**, A. E. Azzaoui, H. Park, D. Camacho, and J. H. Park, "A comprehensive study of quantum computing technologies in smart city: Review and future directions," *Human-centric Computing and Information Sciences*, vol. 14, 2024, Impact Factor: 3.9, JCR Q2 Top 26.3%.
2. S. Jeremiah, **H. Chen**, S. Gritzalis, and J. H. Park, "Leveraging application permissions and network traffic attributes for android ransomware detection," *Journal of Network and Computer Applications*, vol. 230, p. 103 950, 2024, Impact Factor: 7.7, JCR Q1 Top 4.2%.
3. **H. Chen**, A. Azzaoui, S. Jeremiah, and J. H. Park, "A novel smart contract based optimized cloud selection framework for efficient multi-party computation," *Journal of Information Processing Systems*, vol. 19, no. 2, pp. 240–257, 2023, Scopus.
4. **H. Chen**, S. Jeremiah, C. Lee, and J. H. Park, "A digital twin-based heuristic multi-cooperation scheduling framework for smart manufacturing in iiot environment," *Applied Sciences*, vol. 13, no. 3, p. 1440, 2023, Impact Factor: 2.5, JCR Q2 Top 24.3%.
5. A. Azzaoui, **H. Chen**, S. Kim, Y. Pan, and J. H. Park, "Blockchain-based distributed information hiding framework for data privacy preserving in medical supply chain systems," *Sensors*, vol. 22, no. 4, p. 1371, 2022, Impact Factor: 3.4, JCR Q2 Top 30.9%.

Conference Proceedings

1. **H. Chen** and J. H. Park, "Quantum-powered secure multi-party collaboration framework for smart manufacturing," in *Proceedings of the International Conference on Big Data, IoT, and Cloud Computing (BIC)*, Jeju, Korea, Aug. 2023.
2. A. E. Azzaoui, **H. Chen**, and J. H. Park, "Qaoa: A quantum approximate optimization algorithm for optimal base station selection in 6g networks and beyond," in *Proceedings of the International Conference on Computer Science and its Applications (CSA)*, National University of Laos, Laos, Dec. 2022.
3. **H. Chen**, T. W. Kim, and J. H. Park, "Smart contract-based secure multi-party computation for cloud privacy preserving," in *Proceedings of the World Congress on Information Technology Applications and Services (World IT Congress)*, Jeju, Korea, Feb. 2022.
4. **H. Chen**, T. Kim, J. Park, and J. H. Park, "Optimized node selection method for efficient cloud computing based on smart contracts," in *Proceedings of the Korean Information Processing Society Conference*, vol. 29, 2022, pp. 48–51.
5. **H. Chen** and J. H. Park, "Bdt: Blockchain and digital twins-based secure framework for efficient data processing in smart manufacturing," in *Proceedings of the International Conference on Big Data, IoT, and Cloud Computing (BIC)*, Jeju, Korea, Aug. 2022.
6. **H. Chen** and J. H. Park, "Quantum homomorphic cryptography-based efficient medical imaging architecture for iomt in smart city environments," in *Proceedings of the International Conference on Future Information Technology, Applications and Services (IFIT)*, Seoul, Korea, Oct. 2022.
7. **H. Chen**, T. Kim, J. Park, and J. H. Park, "Secure multi-party computation based on homomorphic encryption for privacy preserving in iot networks," in *Proceedings of the Korea Information Processing Society Conference*, 2021, pp. 189–192.
8. **H. Chen**, T. Kim, and J. H. Park, "Multi-blockchain system based on directed acyclic graph for improving data processing performance," in *Proceedings of the Korean Information Processing Society Conference*, vol. 28, 2021, pp. 25–28.

Books

1. Chahot, *Quantum Security: A New Era in Information Protection*, Chinese. Beijing, China: Publishing House of Electronics Industry, 2024, A Chinese-language technical textbook providing a systematic overview of classical-to-quantum cryptography, quantum key distribution, and the transition from classical to quantum-based security frameworks., ISBN: 978-7-121-48136-9. URL: `https://www.phei.com.cn/module/goods/wssd_content.jsp?bookid=65109`.

References

Academic Mentor-Prof. Laurence T. Yang

FCAE, FEIC, MAE, MRAE, MNAAI, FIEEE

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Course Lecturer-Prof. Dongwann Kang

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